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CE-502 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Advanced Surveying
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Describe the various methods of GPS surveying. Also give a brief account of the applicability and limitations of each technique.
2. Write a brief note on electron magnetic energy.
3. Define:
 - i) Azimuth
 - ii) Nadir
 - iii) Residual error
 - iv) Most probable value
4. Mention the advantages of total station and also discuss the working principles of the same.
5. What is relief displacement? Derive an expression for the relief displacement in a vertical photograph.
6. Determine the azimuth and altitude of a star from the following data:
Declination of star = $8^{\circ} 30' S$
Hour angle of star = 322°
Latitude of observer = $50^{\circ} N$

OR

What is "Total station"? Explain its working with a neat sketch.
7. What do you understand by "Aerial photogrammetry"? Also give its use in civil engineering works.

OR

Determine the LMT at the following place whose longitudes are given, if the Greenwich Civil time was 5h 34m 15s PM on July 4, 1966.
 - i) $70^{\circ}20' E$
 - ii) $70^{\circ}20' W$
 - iii) $112^{\circ}38' 30'' E$
8. Briefly explain any two of the following :
 - a) Photogrammetry
 - b) Remote sensing satellites
 - c) Geographic information system
 - d) Digital Planimeter
